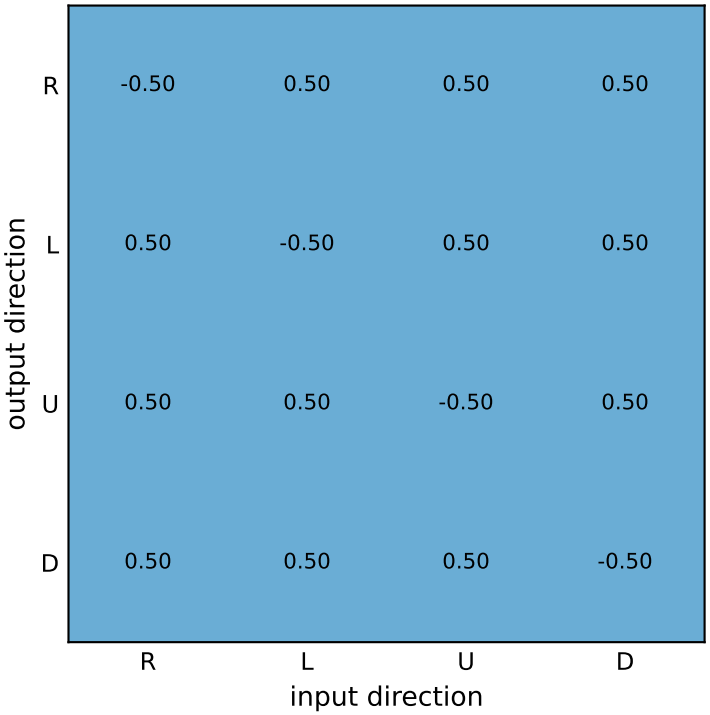
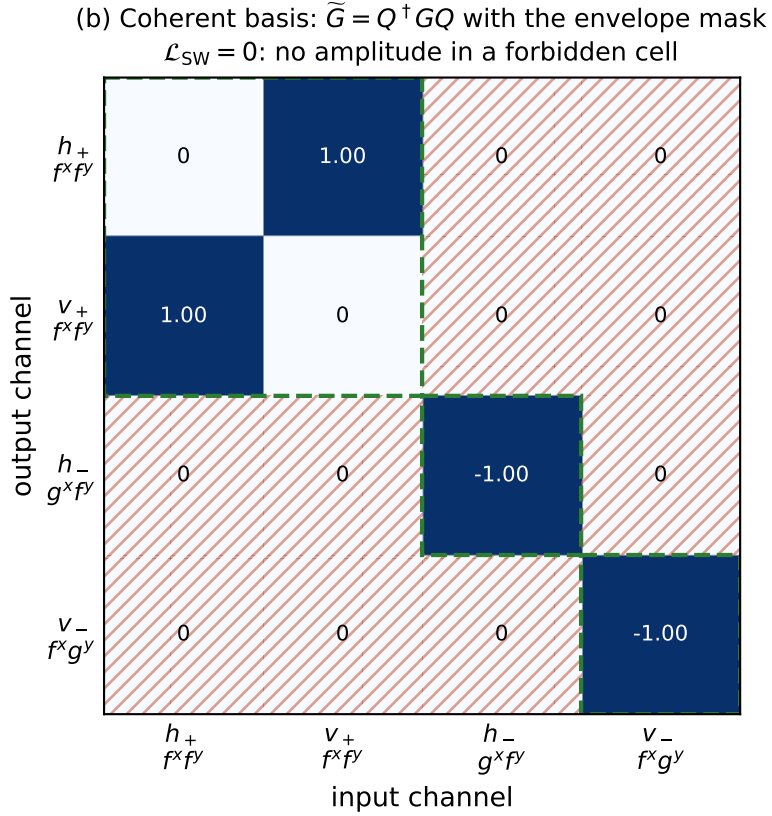

(a) Physical basis: $C_{d'd}$, rows = output d', columns = input d
R
L
U
D
output direction
-0.50
0.50
0.50
0.50
0.50
-0.50
0.50
0.50
0.50
0.50
-0.50
0.50
0.50
0.50
0.50
-0.50
R
L
U
D
input direction


(b) Coherent basis: $\widetilde{G} = Q^{\dagger} G Q$ with the envelope mask
$\mathcal{L}_{\mathrm{SW}} = 0$: no amplitude in a forbidden cell
$h_+ f^x f^y$
$v_+ f^x f^y$
$h_- g^x f^y$
$v_- f^x g^y$
output channel
0
1.00
0
0
1.00
0
0
0
0
0
-1.00
0
0
0
0
-1.00
$h_+ f^x f^y$
$v_+ f^x f^y$
$h_- g^x f^y$
$v_- f^x g^y$
input channel